Pocket Beacon

Design review · 2026-11-08 · EVT-0 release candidate (design complete, prototype not built)



Pocket Beacon is a small item finder that lives on a keyring or inside a bag. When the user cannot find their keys, they tap "Ring" in the phone app, and the beacon plays a loud tone and blinks until it is found.

32 × 32 × 10 mm

Overall size, from the validated CAD verified

13–36 months

On one CR2032, calibrated to baseline est.

$8.89–17.14

Unit cost at 1,000 units, before one-time costs est.

Who uses it, where and how

Who

adults who misplace keys, bags or small cases. Secondary: the same user leaving a bag behind (out-of-range alert in the app).

Moment of use

the user cannot find the keys in the house (the phone rings the beacon), or the user cannot find the phone (a double press on the beacon rings the phone).

Carried or mounted

on a keyring through a slot, or loose in a bag. Pressed against keys, coins and the inside of pockets.

Session pattern

always on. The beacon advertises over BLE all the time. The user rings it 2 to 10 times per month, for 5 s to 30 s each time.

Phone or app

required for ringing and setup. iOS and Android. Standard BLE GATT, with no network membership in v1 (D-003).

Water and sweat

EXPECTED

rain, a drop into a puddle or a sink, sweat from hands

DESIGN ANSWER

IP67 (1 m, 30 min), sealed lid with O-ring, sealed sound port, no open button (S-09)

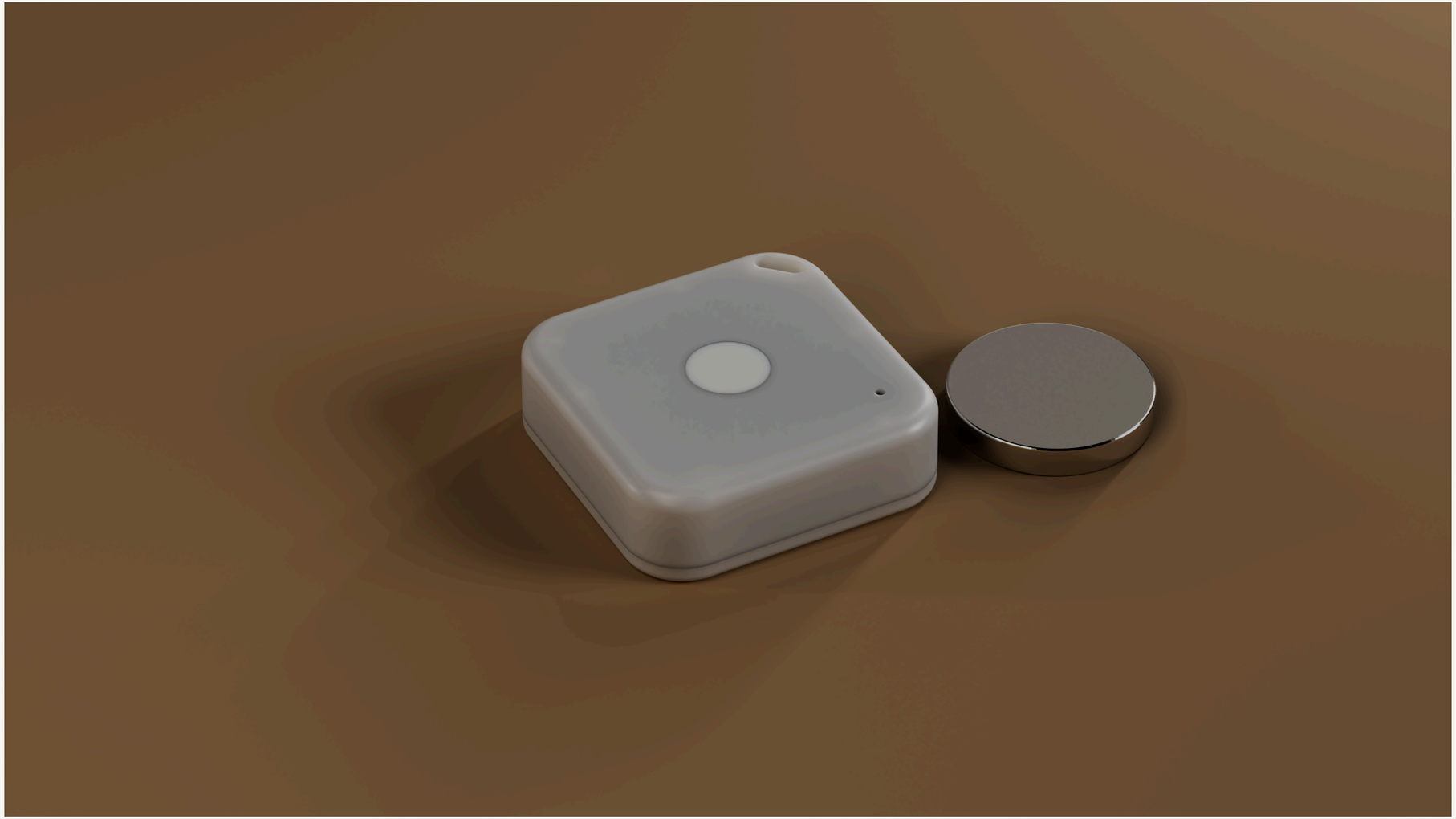
Temperature (use / storage / charging)

EXPECTED

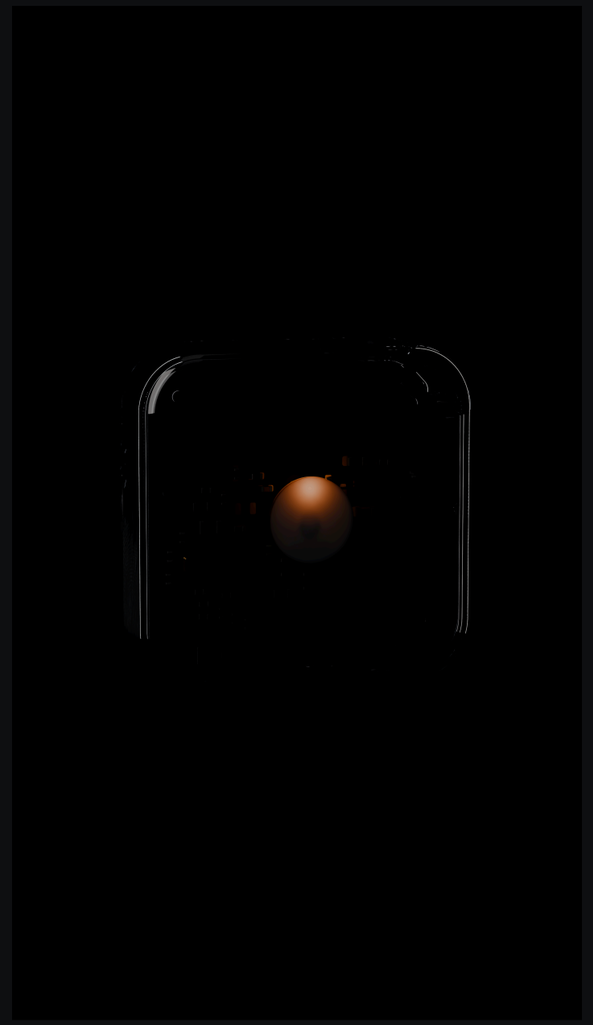
–10 °C to +45 °C use, –20 °C to +60 °C storage (car), no charging

DESIGN ANSWER

parts rated –40 °C to +85 °C; coin-cell capacity drops at –10 °C, so the alarm must still work at the end-of-life voltage (S-06)



Chalk, next to the CR2032 it runs on for more than a year. The cell is user-replaceable behind a screw-secured door.



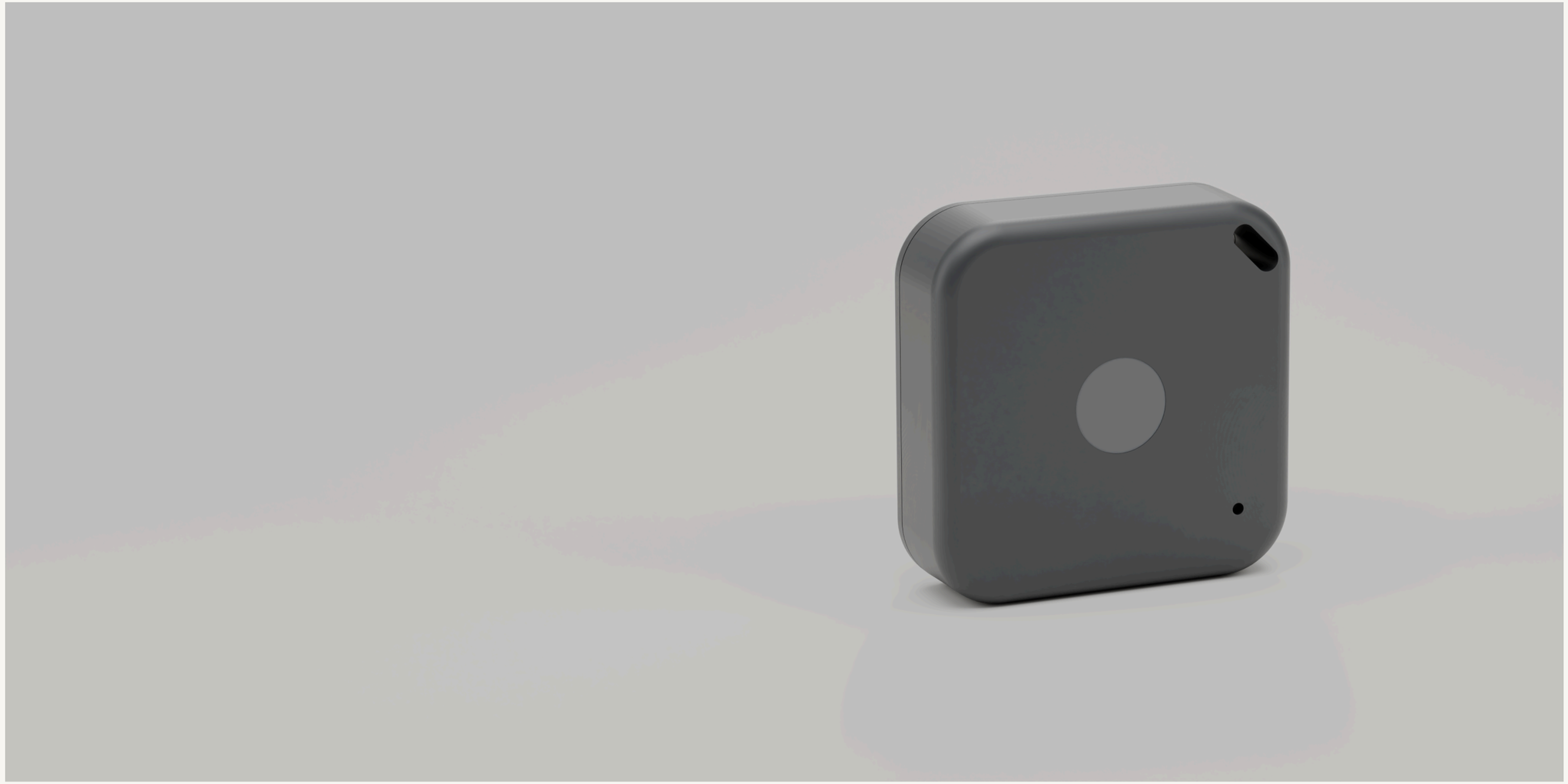
Smoke. The amber status light glows through the translucent button membrane.

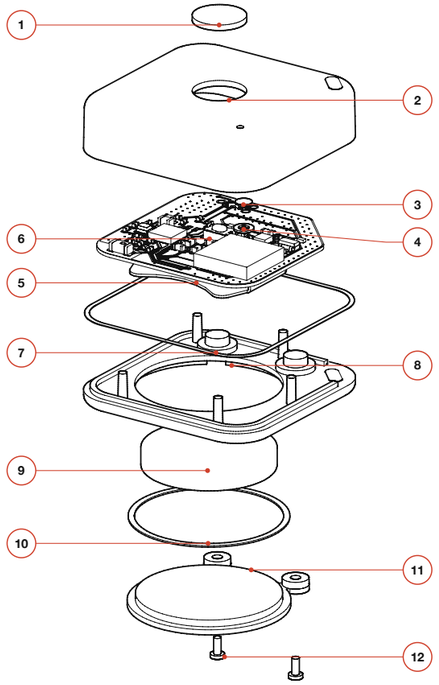


Everything inside, top to bottom: TPU button cap, front shell, acoustic membrane, 4-layer board, CR2032, adhesive ring, rear plate, door gasket, door and screws.



Three resins, one mold: graphite, chalk and smoke.





BILL OF MATERIALS

Item	Description	Specification	Qty
1	**Button**	TPU 80A translucent, 2-shot molded into the shell; light path for status LED D1; envelope 8.0 × 8.0 × 1.1 mm	1
2	**Front shell**	PC/ABS, injection molded, 0.9 mm wall, 1° draft, R6 corners; vertical keyring slot 2.0 × 5.0 mm through the +X+Y corner; Ø1.0 sound hole; envelope 32.0 × 32.0 × 8.1 mm	1
3	**Acoustic vent**	ePTFE membrane, die-cut Ø2.5, PSA backed, over the sound hole; envelope 2.5 × 2.5 × 0.1 mm	1
4	**Port gasket**	Soft PU foam, die-cut 3.0 × 3.0, 0.8 mm free height; seals the LS1 top port to the shell; envelope 3.0 × 3.0 × 0.3 mm	1
5	**CR2032 retainer**	Keystone 3034TR, SMD, on the PCBA back side (LCSC C5213768); envelope 23.2 × 16.4 × 4.0 mm	1
6	**Main board (PCBA)**	FR4 4 layers, 0.6 mm, ENIG, black mask; CC2340R52 BLE SoC, PAM8904 + CPT-9019A sounder; JLCPCB assembly; envelope 28.4 × 28.4 × 2.7 mm	1
7	**PSA ring**	Die-cut acrylic PSA, 0.10 mm; bonds the back plate to the shell; envelope 31.7 × 31.7 × 0.1 mm	1
8	**Back plate**	PC/ABS, shell color, injection molded; 2 blind screw bosses; laser-marked regulatory text; envelope 32.0 × 32.0 × 6.0 mm	1
9	**Coin cell**	CR2032 lithium 3 V, Ø20.0 × 3.2 mm (Panasonic CR2032); user replaceable; envelope 20.0 × 20.0 × 3.2 mm	1
10	**Door gasket**	Silicone 50A, flat die-cut, 0.23 mm compressed; IP67 face seal; envelope 23.7 × 23.7 × 0.2 mm	1
11	**Cell door**	PC/ABS, shell color; tool-secured per UL 4200A; laser-marked warning icons; envelope 26.1 × 24.0 × 1.9 mm	1
12	**Captive screw**	Thread-forming 1.4 × 3 mm, stainless A2, captive in the door	2

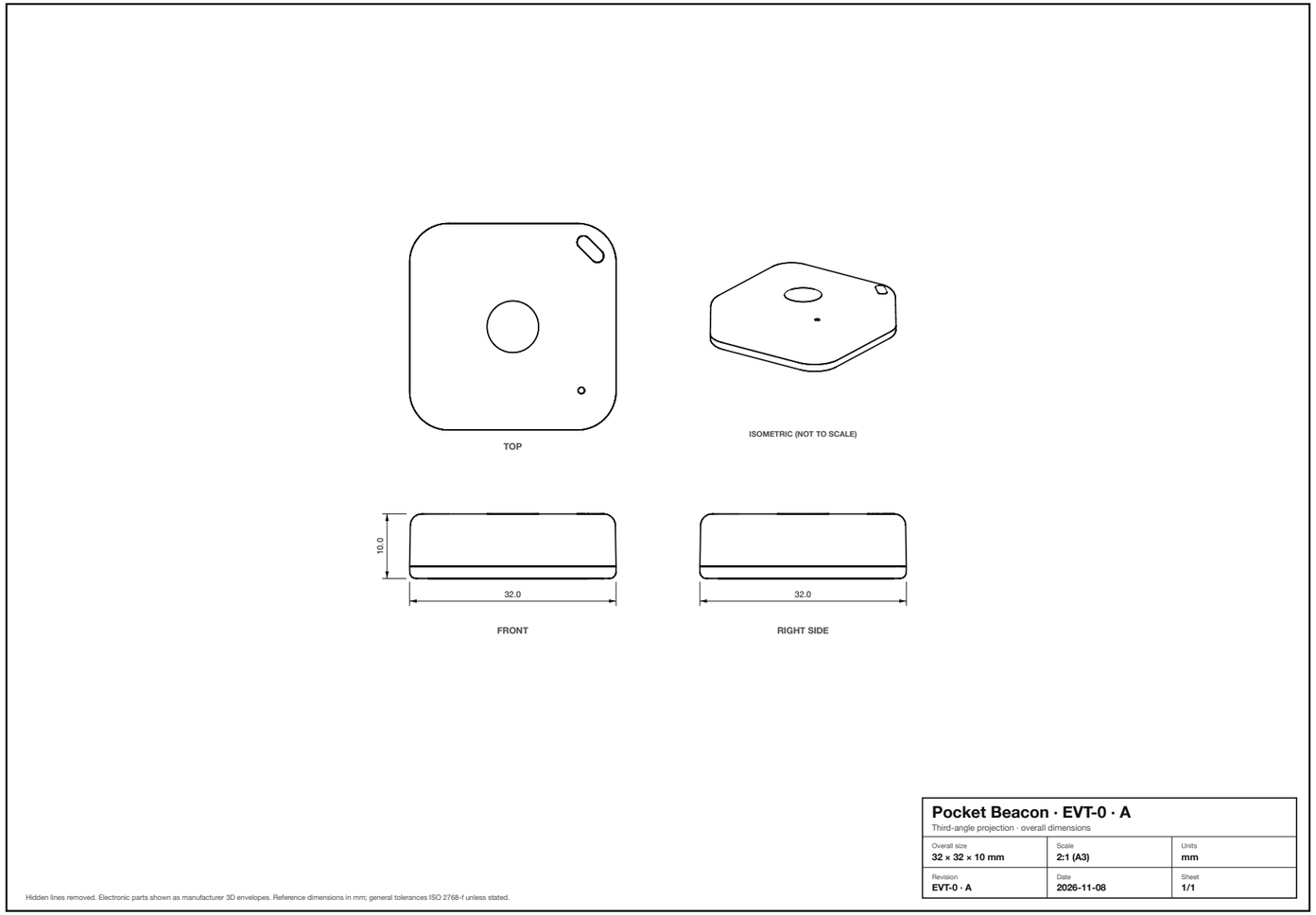
Hidden lines removed. Electronic parts shown as manufacturer 3D envelopes. Reference dimensions in mm; general tolerances ISO 2768-f unless stated.

Pocket Beacon · EVT-0 · A

Exploded isometric view · assembly for quotation

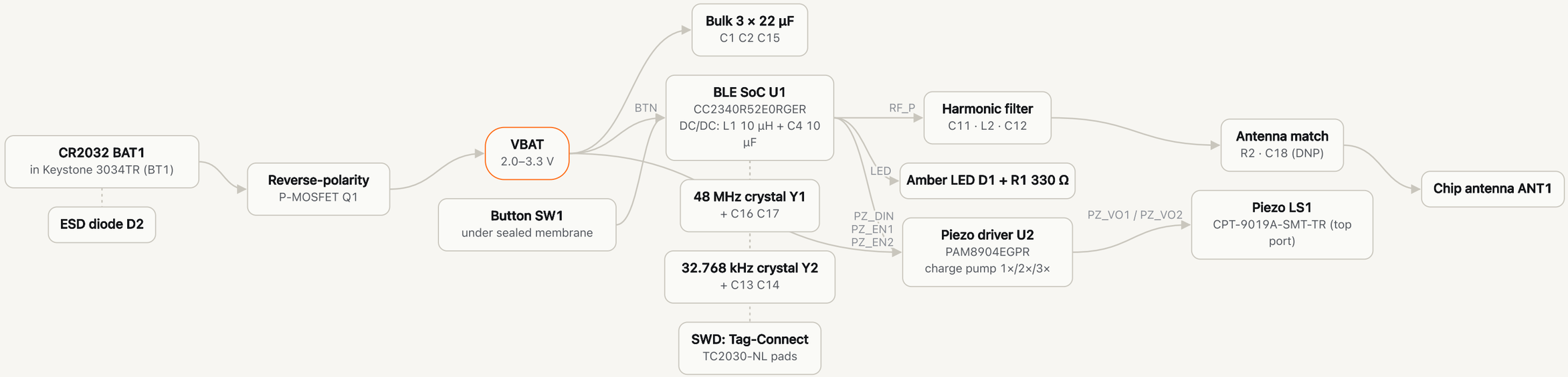
Overall size	Scale	Units
32 × 32 × 10 mm	**2:1 (A3)**	**mm**
Revision	Date	Sheet
EVT-0 · A	**2026-11-08**	**1/1**

Exploded view for quotation: 12 items, from the button down to the two captive screws, with the round CR2032 between the back plate and the door.



Overall size 32 × 32 × 10 mm, third-angle projection, scale 2:1 on A3.

What is inside



Rail data for the scripts is in `hw/power/rails.json` (owner `battery-and-power`). VBAT `max_ma` = 15 mA is the planning limit for **continuous** load on a CR2032 at room temperature (F-204). Short TX bursts above it are carried by the bulk capacitance (§4).

Components 1/6

BLE 5.3 SoC with integrated DC/DC

U1

PART

CC2340R52E0RGER

Texas Instruments · VQFN-24 4x4 0.5 mm pitch (RGE)

1 PC

$1.49

100

$0.94

1K

$0.84

WHERE TO BUY

LCSC · DigiKey · TI

ALTERNATIVES

CC2340R52N0RGER
CC2340R53E0RKPR

verified

Piezo sounder driver with charge pump (1x/2x/3x)

U2

PART

PAM8904EGPR

Diodes Incorporated · W-QFN2020-12 (Type US)

1 PC

$0.49

100

$0.28

Components 2/6

RSTN pull-up 100 kohm

R3

PART

0402WGF1003TCE

UNI-ROYAL (Uniroyal Electronics) · 0402

1 PC

$0.001

100

$0.001

1K

$0.001

WHERE TO BUY

LCSC · JLCPCB

ALTERNATIVES

RC0402FR-07100KL

est.

Tactile switch under sealed membrane

SW1

PART

KMR221GLFS

C&K · SMD 4.2 x 2.8 mm, 4 terminals + ground pin

1 PC

$0.67

100

$0.48

1K

Components 3/6

ESD protection on cell holder (VBAT)

D2

PART

TPD1E10B06DPYR

Texas Instruments · X1SON (DPY) 2-pin, 1.0 x 0.6 mm

1 PC

$0.41

100

$0.16

1K

$0.12

WHERE TO BUY

DigiKey · LCSC

ALTERNATIVES

TPD1E10B06DYAR

ESD9B5.0ST5G

verified

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48 MHz crystal, 7 pF, -20 to 70 C

Y1

PART

ABM11W-48.0000MHZ-7-B1U-T3

Abracon · SMD 2016-4P

1 PC

$0.54

100

$0.31

Components 4/6

Bulk capacitor 22 uF X5R 6.3 V for coin-cell pulses

C1,C2,C15 × 3

PART

CL10A226MQ8NRNC

Samsung Electro-Mechanics · 0603 (1608 metric) MLCC

1 PC

$0.017

100

$0.017

1K

$0.011

WHERE TO BUY

LCSC · DigiKey JLC basic

ALTERNATIVES

CL21A226MAQNNNE

GRM188R60J226MEA0D

+1 more in the dossier

verified

VDDS decoupling 10 uF X5R

C3

PART

CL10A106KP8NNNC

Samsung Electro-Mechanics · 0603 (1608 metric) MLCC

1 PC

$0.02

100

$0.02

Components 5/6

100 nF X7R decoupling: VDDS pins 11/18/24 (C5, C6, C21), VDDR pins 2/20 (C22, C23), RSTN (C20)

C5,C6,C20,C21,C22,C23 × 6

PART

CL05B104KO5NNNC

Samsung Electro-Mechanics · 0402 (1005 metric) MLCC

1 PC

$0.003

100

$0.003

1K

$0.002

WHERE TO BUY

LCSC JLC basic

ALTERNATIVES

GRM155R71C104KA88D

CL05A104KA5NNNC

verified

U2 CIN and COUT 2.2 uF 16 V

C7,C8 × 2

PART

CL10A225KO8NNNC

Samsung Electro-Mechanics · 0603 (1608 metric) MLCC

1 PC

$0.012

100

$0.012

Components 6/6

RF harmonic filter series inductor 2.7 nH (start, tune on EVT)

L2

PART

LQG15HS2N7B02D

Murata Electronics · 0402

1 PC

$0.062

100

$0.053

1K

$0.045

WHERE TO BUY

LCSC · JLCPCB

ALTERNATIVES

LQG15HS2N7S02D

LQP15MN2N7B02D

est.

Antenna match series slot 2: 0 ohm for the pi match

R2

PART

0402WGF0000TCE

UNI-ROYAL (Uniroyal Electronics) · 0402

1 PC

$0.001

100

$0.001

Baseline lasts 3.0 years on a 225 mAh cell

3 V · 85% usable · range: end of life to new cell · not measured yet est.



- **Within limit** REQ-005 limit, cell current (<= 20 mA): worst-case 15.1 mA against 20 mA
- **Within limit** CR2032 pulse capability used for planning (30 mA peaks cost about 9 % capacity vs 15 mA): worst-case 15.1 mA against 30 mA

Ready to build · prototype

Maturity: EVT-0 release candidate (design complete, prototype not built)

BY AREA

Parts	Go
Electrical	Go
Power and battery	Go
PCB	Go
Mechanical	Go
Compliance	Go
Cost	Go
Schedule	Go
Claims vs reality	Go

BUILD PACKAGE

Done	Routed boards	release evt0-a
Done	DRC and ERC at 0	release evt0-a
Done	Fab and assembly files	release evt0-a
Done	BOM in stock	release evt0-a
Done	Enclosure STEP	release evt0-a

Prototype checks 1/3

Each check has a method and a pass threshold. None of them blocks building the prototype.

Standby floor and weighted daily average current (power.md M1, M6)

METHOD AND INSTRUMENT

bench supply 3.0 V at TP1/TP2, cell removed; 60 s idle, then the 24 h use-pattern script with a real phone · power profiler (Nordic PPK2 or Joulescope), ≥ 100 kHz sampling

PASS

standby ≤ 1.2 µA; daily average ≤ 14 µA (21.2 µA limit, REQ-002)

Charge per slow advertising event and per connection event (M2, M3)

METHOD AND INSTRUMENT

3.0 V source, 0 dBm, 31-byte payload, 100 events averaged; 1000 ms connection, empty packets · power profiler (Nordic PPK2 or Joulescope), ≥ 100 kHz sampling

PASS

≤ 12 µC per advertising event (else 2000 ms interval); ≤ 4.5 µC per connection event

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Ring current in 2× and 3× and the adaptive ring rule (M4, D-028)

METHOD AND INSTRUMENT

source 3.0 V and 2.5 V; ring command from the app; log PZ_EN1/PZ_EN2 and VBAT · power profiler (Nordic PPK2 or Joulescope), ≥ 100 kHz sampling + logic analyzer

PASS

3× ≤ 15.5 mA, 2× ≤ 7.5 mA; firmware selects 2× at 2.5 V under load and 3× at 3.0 V

Ring loudness (REQ-006)

METHOD AND INSTRUMENT

assembled unit hanging free, mic at 25 cm on axis, 4 kHz ring, fresh cell and a cell at 2.4 V under load · class 2 sound level meter, A-weighted, IEC 61672

PASS

≥ 75 dB(A) in 3× on a fresh cell; ≥ 65 dB(A) in 2× at 2.4 V loaded

Prototype checks 2/3

Each check has a method and a pass threshold. None of them blocks building the prototype.

32.768 kHz start-up at cold with ESR 90 kΩ (D-025, F-105)

METHOD AND INSTRUMENT

10 power cycles at –10 °C; measure LFXT start time and sleep-clock accuracy · oscilloscope + TI SDK clock test

PASS

starts every time within 1 s; accuracy ≤ ±50 ppm

Loaded cell voltage at cold and brown-out margin (M7, M8)

METHOD AND INSTRUMENT

CR2032 discharged to 50 % (15 kΩ for 680 h), –10 °C chamber, 2× ring; fast advertising through 1 kΩ from 2.7 V · power profiler (Nordic PPK2 or Joulescope), ≥ 100 kHz sampling in ammeter mode + climate chamber

PASS

VBAT ≥ 2.3 V during the ring; VBAT ≥ 2.0 V and no reset over 30 s

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BLE range through a bag (REQ-010)

METHOD AND INSTRUMENT

phone app RSSI and connection at 5 m and 10 m indoor, unit inside a fabric bag with keys · phone app log

PASS

connected at 10 m indoor (≥ 5 m limit)

Sound port alignment over the LS1 top port (Q-CD-01)

METHOD AND INSTRUMENT

view the assembled front shell hole against LS1 under a stereo microscope; then V-004 · stereo microscope with reticle

PASS

port centre within 0.3 mm of the Ø1.0 shell hole centre

Prototype checks 3/3

Each check has a method and a pass threshold. None of them blocks building the prototype.

Ingress seal of the door and port gaskets (REQ IP67 target)

METHOD AND INSTRUMENT

assembled EVT unit (machined or molded parts; SLA parts only for a pre-check), immersion 1 m for 30 min per IEC 60529 IPX7, then open and inspect · immersion tank, moisture indicator paper inside

PASS

no water inside, indicator unchanged, unit still advertises

Child-resistant compartment pre-test (UL 4200A securement)

METHOD AND INSTRUMENT

door closed with both screws at 0.08 N·m; apply the UL 4200A abuse sequence on 3 units (drop, crush, torque on the door) · torque driver, drop rig 1 m, force gauge

PASS

door stays secured and the cell stays inaccessible on all 3 units

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Button press force through the TPU membrane

METHOD AND INSTRUMENT

press the button cap centre at 1 mm/s, 10 presses · force gauge with ≥ 0.01 N resolution

PASS

2.3–3.3 N actuation force, tactile click felt, no pre-press at rest

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Open findings

Minor

FINDING

Y2 ESR 90 kΩ leaves 10 % margin to the CC2340 100 kΩ limit

Parts

CHANGE

—

OWNER

specs-engineer

Minor

FINDING

Advertising and connection event charges are estimates

Power and battery

CHANGE

—

OWNER

battery-and-power

Minor

FINDING

48 MHz external load capacitors: the TI LaunchPad leaves them unfitted, the datasheet requires them for compliance

Electrical

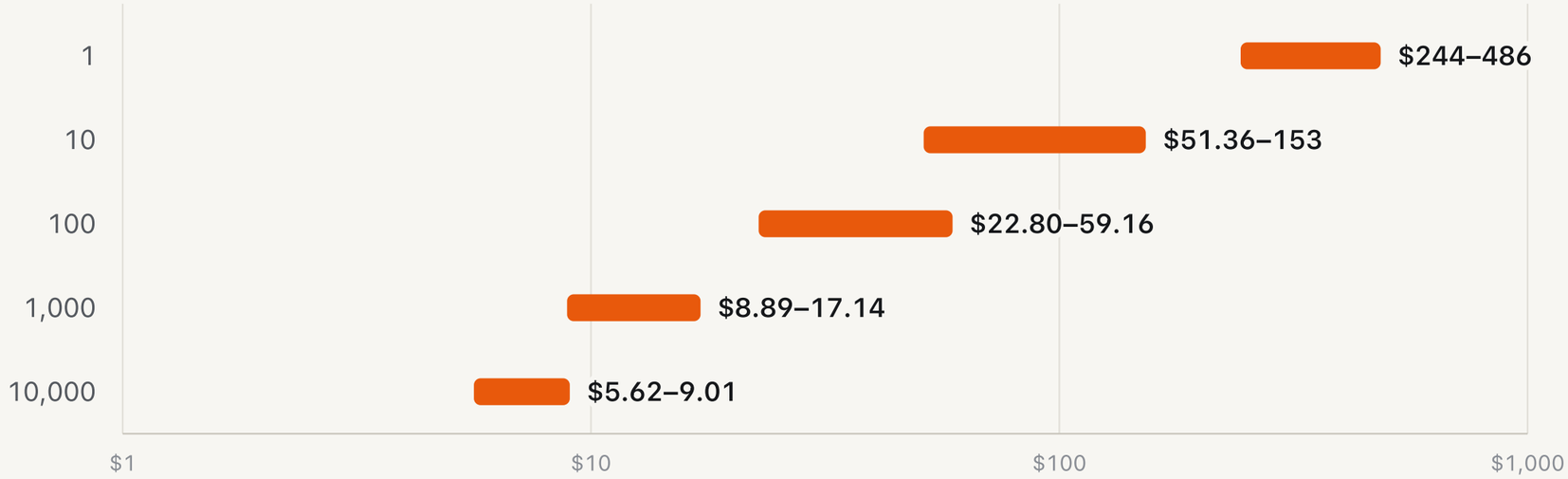
CHANGE

—

OWNER

pcb-engineer

$8.89–17.14 per unit at 1,000 units



Per unit, before one-time costs. Ranges are low–high, USD, 2026-11-08. Logarithmic axis; units per order on the left. 39% of the 1,000-unit cost rests on verified prices.

$50.74–174 per unit at 1,000, one-time costs included

1

PER UNIT
$244–486

ONE-TIME, PER UNIT
$80.00–300

TOTAL PER UNIT
$324–786

LANDED, US (FORMAL ENTRY, COMPANY IMPORT)
$344–1,062

10

PER UNIT
$51.36–153

ONE-TIME, PER UNIT
$8.00–30.00

TOTAL PER UNIT
$59.36–183

LANDED, US (FORMAL ENTRY, COMPANY IMPORT)
$63.36–266

100

PER UNIT
$22.80–59.16

ONE-TIME, PER UNIT
$258–637

$41,850–156,800 one-time to reach 1,000 units

Certification added for the 1k run: EU (RED radio + EN 18031 + RF exposure + CRA + GPSR + WEEE + batteries + packaging), UK (UKCA, PSTI, WEEE/batteries), Brazil (Anatel + take-back)

GROUP
certification

COST
$12,750–82,000

est.

Certification for the launch batch, sold in the US only: FCC 15C + 15B, Bluetooth SIG, UL 4200A report, IP67 test, pre-scan, UN 38.3 summary, DULT firmware, RoHS/Prop 65 declarations

GROUP
certification

COST
$20,300–46,500

est.

Rapid aluminum T1 tools, shell + lid, textured

GROUP
tooling

COST
$5,000–12,000

est.

Test and programming fixture (EMS-built, RF + current + ring)

GROUP
fixtures

COST

Certifications

All markets

COST (USD)
$13,300–22,000

LONGEST LEAD (WEEKS)
4

United States

COST (USD)
$7,000–24,500

LONGEST LEAD (WEEKS)
6

European Union

COST (USD)
$9,650–68,700

LONGEST LEAD (WEEKS)
8

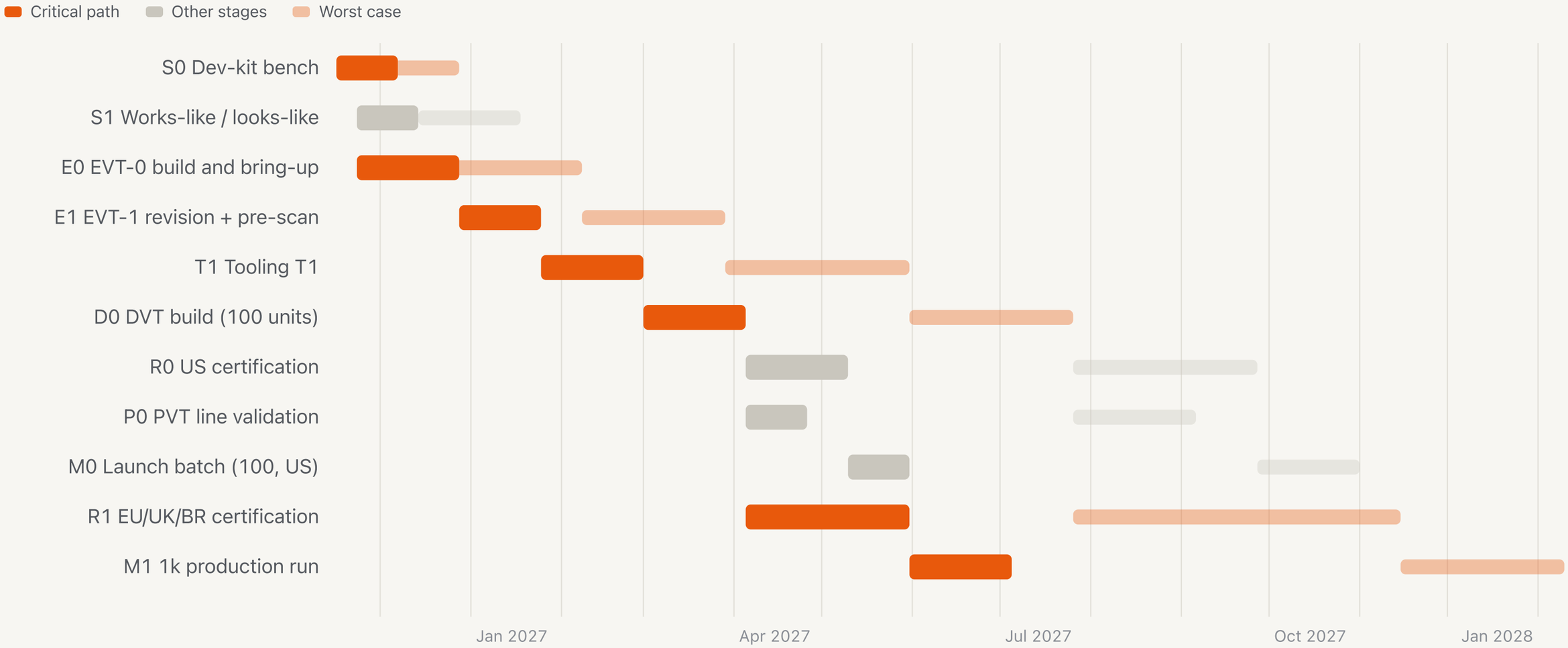
United Kingdom

COST (USD)
$500–20,500

LONGEST LEAD (WEEKS)
4

Brazil

Roadmap: Jul 2027 (best) to Jan 2028 (worst)



Engineering freeze
2027-01-25 → 2027-03-29

First legal sale (US)
2027-05-10 → 2027-09-27

First legal sale in Brazil
2027-05-31 → 2027-11-15

How to read the numbers

Components: 20 verified · 7 estimated · 0 assumed

verified

Read from a named source in this review, with its date.

est.

Calculated or inferred; the basis is stated next to the number.

assumed

Chosen for planning. A measurement or a quote must replace it.